POWER MODULE

ARK 12S Payload Power Module

Dual 5.2 V / 6 A + 12 V / 6 A regulated supply and digital power monitor for the Pixhawk Autopilot Bus

The USA-built ARK 12S Payload Power Module provides dual regulated outputs and high-accuracy digital battery monitoring for Pixhawk Autopilot Bus carriers. A 100 A continuous battery path is monitored by a TI INA238 while two independent synchronous bucks deliver 5.2 V / 6 A for avionics and 12 V / 6 A for payloads.

The 6-pin Molex output matches the ARK PAB Carrier power pinout; the additional 4-pin Molex supplies the 12 V payload rail. Parallel modules increase continuous battery current beyond 100 A.

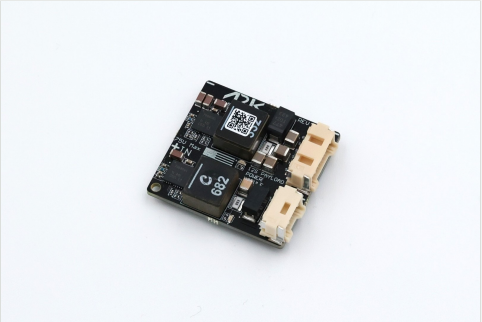
KEY FEATURES

- 100 A continuous battery current path (parallel for higher)
- Dual TI LM70880 bucks — 5.2 V / 6 A + 12 V / 6 A
- TI INA238 digital power monitor with 0.1 mΩ shunt (I²C)
- 10–75 V input (15 V min for 12 V rail at full load)
- 6-pin Molex (5 V / I²C) matching PAB pinout
- 4-pin Molex for 12 V payload
- Includes both Molex cables

SPECIFICATIONS

ELECTRICAL	
Input voltage	10–75 V (15 V min @ 12 V / 6 A)
Battery current	100 A continuous (20 °C ambient)
5.2 V output	6 A continuous
12 V output	6 A continuous
Power monitor	INA238, 0.1 mΩ shunt, I²C (default 0x45)
Operating temperature	−40 °C to +85 °C
Storage temperature	−40 °C to +85 °C

MECHANICAL	
Dimensions	37 × 35 × 13 mm
Weight	20.5 g



AT A GLANCE	
Input	10–75 V
Battery current	100 A cont.
5.2 V rail	6 A
12 V rail	6 A
Monitor	INA238 0.1 mΩ
Dimensions	37 × 35 × 13 mm
Weight	20.5 g
Hardware rev	Rev 1.2

COMPLIANCE & SOURCING

- Designed and manufactured in the USA
- NDAA compliant

ORDERING & RESOURCES

STORE arkelectron.com/product/ark-12s-payload-power-module/

DOCS docs.arkelectron.com/products/power/ark-12s-payload-power-module

CAD github.com/ARK-Electronics/ark-hardware

5 V / I²C CONNECTOR PINOUT

PIN	SIGNAL	VOLTAGE	NOTES
1	VBRICK	5.2 V	Regulated avionics output, 3 A per pin
2	VBRICK	5.2 V	Regulated avionics output, 3 A per pin
3	SCL	3.3 V	I²C clock to INA238
4	SDA	3.3 V	I²C data to INA238
5	GND	GND	
6	GND	GND	

Connector: Molex 5024940670 6-position CLIK-Mate. Matches ARK PAB Carrier power pinout.

12 V PAYLOAD CONNECTOR PINOUT

PIN	SIGNAL	VOLTAGE	NOTES
1	12 V	12.0 V	Regulated payload output, 3 A per pin
2	12 V	12.0 V	Regulated payload output, 3 A per pin
3	GND	GND	
4	GND	GND	

Connector: Molex 5024940470 4-position CLIK-Mate.

NOTES

- The power monitor measures only current flowing from the battery input pads to the battery output pads. Wire the ESCs and every other battery load to the output pads.
- Each output has its own short-circuit and thermal protection and recovers on its own. The power monitor runs from the 5.2 V output, so a 5.2 V fault also stops battery monitoring.
- Active cooling is recommended when operating at high battery current or high 5 V / 12 V regulator loads.
- For continuous battery current above 100 A, operate multiple modules in parallel.
- Battery input and output are solder pads for high-current connections.

ALTERNATIVES

PRODUCT	KEY DIFFERENCE	WHEN TO CHOOSE
ARK 12S PAB Power Module	5.2 V only (no 12 V rail)	Avionics power only, smaller size / lower weight
ARK PAB Power Module	Lower voltage / 60 A path	6S systems

ARK-DS-ARK-12S-Payload-Power-Module | Version 1.0, 2026-10-29 | Compiled from the ARK 12S Payload Power Module Production schematic Rev 1.2 (25 Aug 2026), the Production bill of materials, the manufacturing documentation, and ARK Electronics published documentation. Temperature ratings per component data sheets. Specifications are subject to change without notice. © 2026 ARK Electronics LLC.